Title

Digital Twin for Liver Resection Planning

Institution

Université Clermont Auvergne

Internship place

ISPR / Institut Pascal, Clermont-Ferrand, France

Supervisors name

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Brief Description of the project

Introduction

Liver resection is a complex surgical procedure that requires careful planning to remove tumors while preserving as much healthy liver tissue as possible. Surgeons currently rely mainly on preoperative medical images and 3D reconstructions to understand the patient's anatomy and plan the surgical strategy.

This internship aims to develop a patient-specific digital twin of the liver and surgical environment to assist surgeons in planning liver resection procedures before surgery. The digital twin will combine medical imaging, 3D anatomical reconstruction, and surgical knowledge to create an interactive virtual representation of the patient's anatomy.

The system will allow surgeons to visualize the liver, tumors, and vascular structures and investigate different resection strategies before performing the procedure.

Main research axes include:

- Digital twin construction: developing an interactive virtual representation of the patient's liver and relevant surgical anatomy.
- Surgical planning: allowing surgeons to define and simulate different resection planes and strategies.
- Surgical simulation: investigating the potential effects of a planned resection on the remaining liver volume.
- Interactive visualization: developing an intuitive interface allowing surgeons to explore and modify the surgical plan.

Expected outcomes

Successful outputs of this internship will lead to a scientific publication in one of the robotics conferences which is a plus for the intern for his/her future academic career.

Skills

Coding : Python,
Knowledge : linear algebra, robotic control, computer vision, 3D geometry, deep learning.